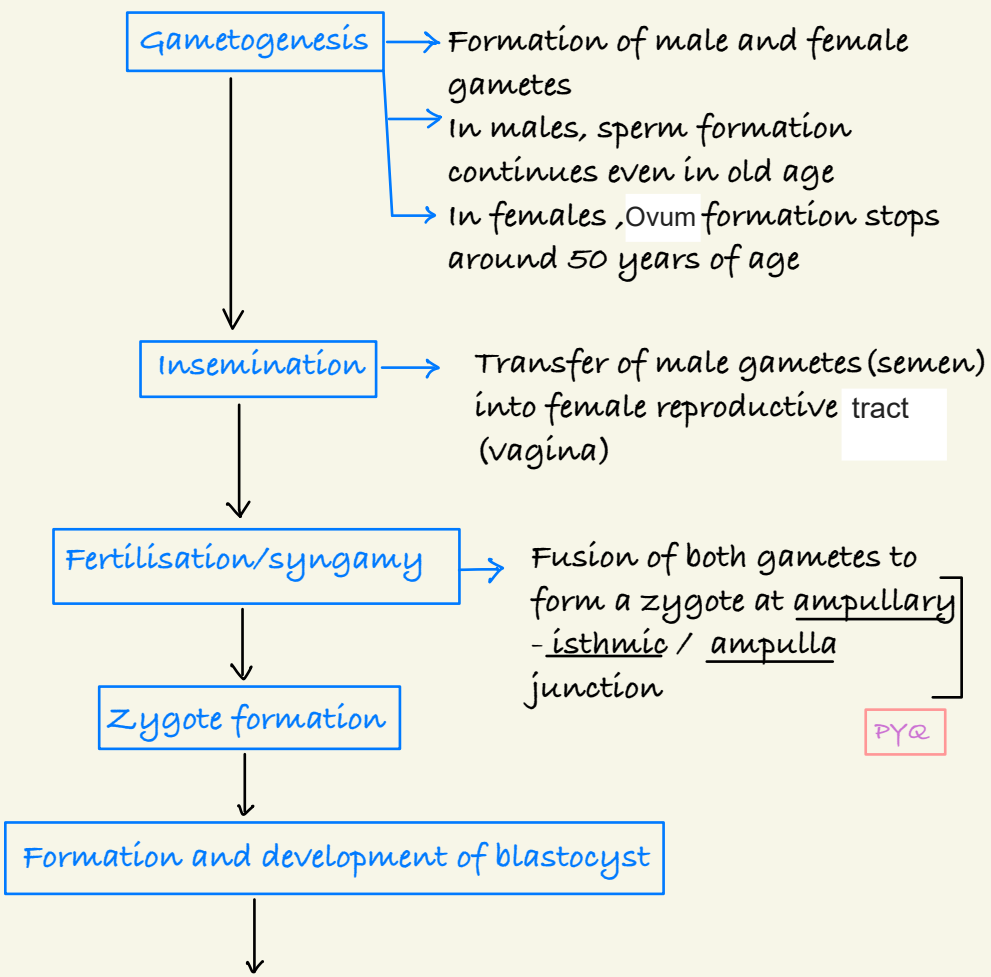
12th hacker ka jugaad

Chapter: Human reproduction

Question) why are human testes located outside the abdominal cavity? Name the pouch in which they are present.

Solution) human testes are located outside the abdominal cavity as it helps in maintaining low temperature (2- 2.5 degree Celsius lower than the body temperature) required for spermatogenesis

Events of human reproduction (3/4 marks question)



Implantation → Attachment of Blastocyst into uterine wall

↓

Gestation → embryonic development (human = 9 months 7 days)

↓

Parturition/ child birth/ delivery

↓

Lactation/ breastfeeding

Male reproduction system (4/5 marks questions)

Primary sex organ

↓

Testis

- produce male gametes
- Produce male sex hormones

(androgens- group of steroidal hormones)

↓

Most common- testosterone

Secondary sex organ

Maintains function of male accessory glands and ducts

Secondary sex organs

External genitalia.

1. Penis.
2. Scrotum.

Accessory gland.

1. Seminal vesicle (2)
2. Prostate. (1)
3. Bulbo- urethral/ cowper's (2)

Accessory duct.

Inside Testis

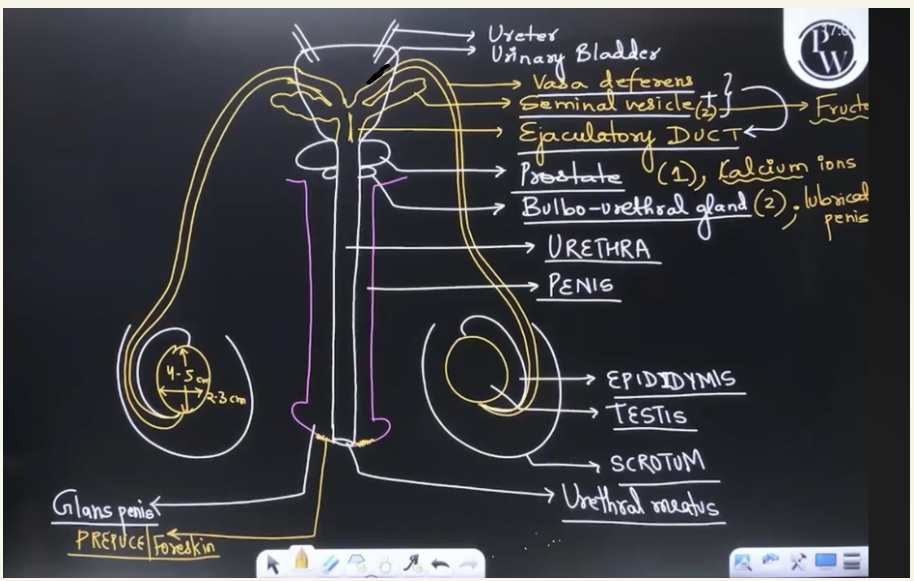
Rete- testis

↓

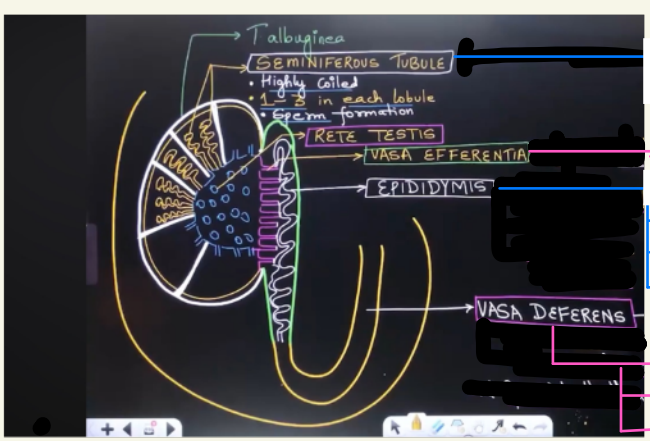
Vasa efferentia / efferent ductules

Note: these ducts store and transport Sperms to outside (from testis) through urethra

Epididymis
↓
Vas deferens
↓
ejaculatory duct
↓
urethra



Structure of testis

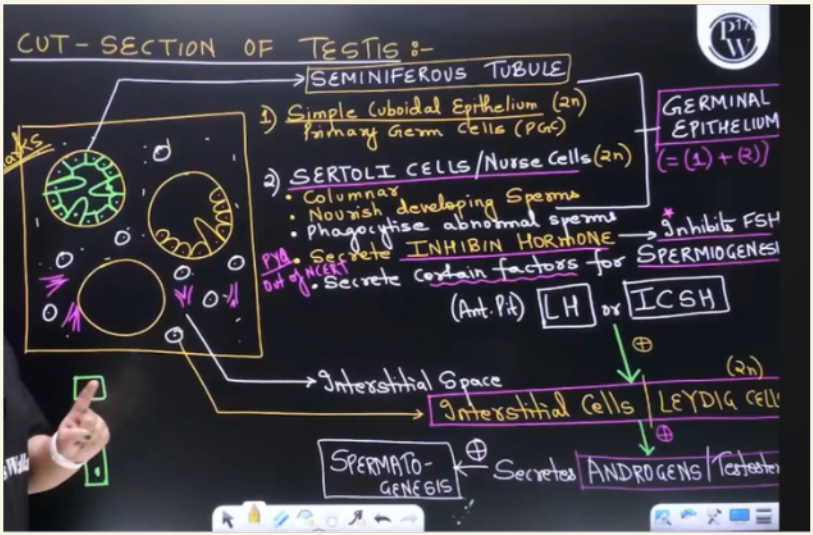


Leave testis and opens into epididymis

- Highly coiled
- Posterior to testis
- Storage and physiological maturation of sperms

- Ascends to abdomen
- Loops over urinary bladder
- Receives a duct from seminal vesicle and opens into urethra as ejaculatory duct

Cut section of testis/ seminiferous to be (2/3 marks)



Hormonal regulation in male

At puberty: GnRH (hypothalamus)

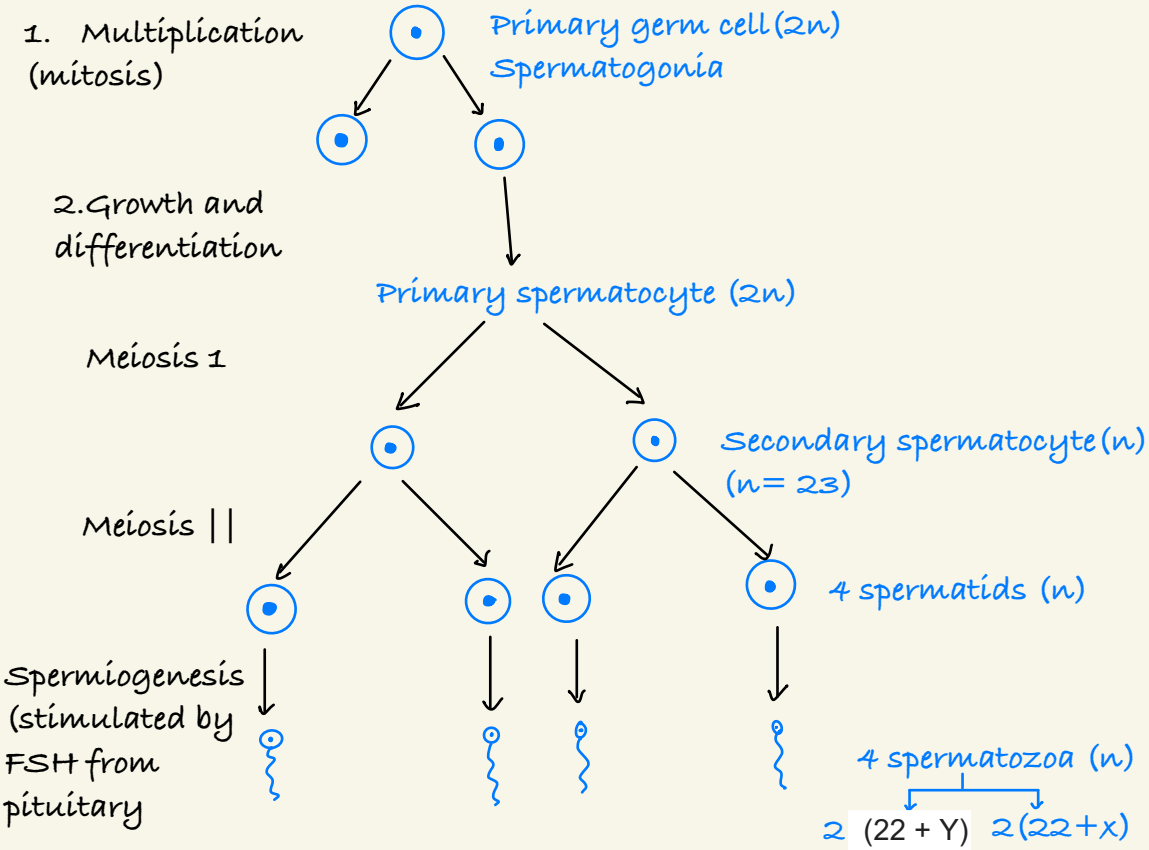
↓

Anterior pituitary (Gonadotropins)

FSH. → Sertoli cells → Inhibin. / Certain factors. → Spermiogenesis (spermatid —> sperm) Transformation

LH./ICSH. → Leydig cells → Androgen/ testosterone → Spermatogenesis

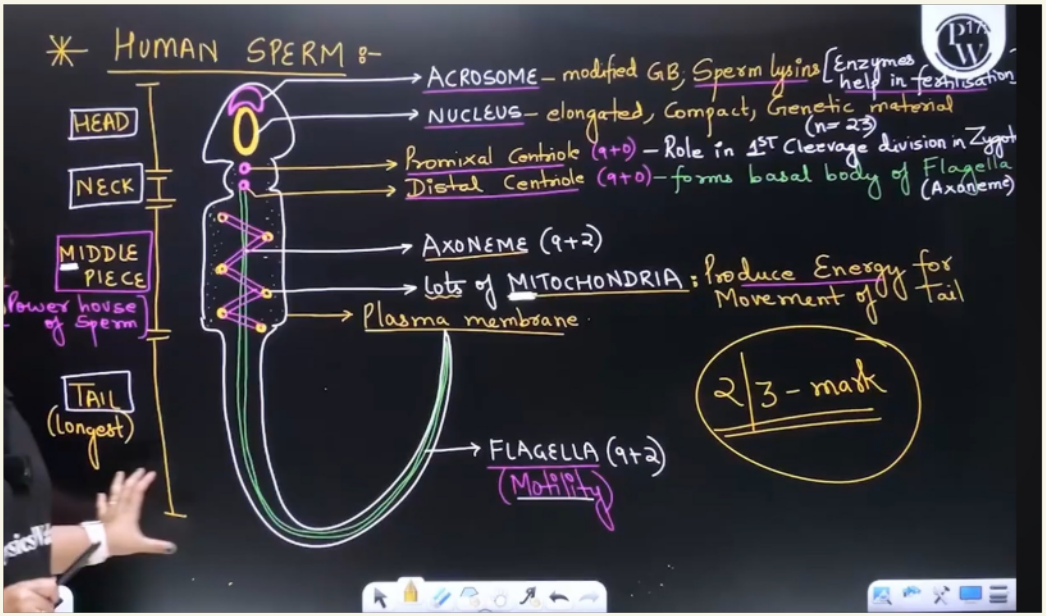
Spermatogenesis



spermiation

Sperms embedded in Sertoli cells, detach from Sertoli cells, and released into lumen of seminiferous tubule (pyq)

Structure of human sperm (2/3 marks question)



Question

Write the location and functions of following in human testes **(2014 , 2 marks)**

(i) Sertoli cells

(ii) Leydig cells cells

Or

Name the cells that nourish the germ cells in the testes where are these cells located the testes?

Sol.

(i) Location of Sertoli cells :Within the lining of seminiferous tubule of testis.
Function of sertoli cells : They provide nutrition to the developing sperms or germ cells

(ii) Location of Leyding cells : In the interstitial spaces between the seminiferous tubules.

Function of Leyding cells : They synthesise and secrete male hormones, ie, androgens testos

Question

Write the function of the seminal vesicle. / Prostate / Cowpers.

Sol.

inal vesicle produces and alkaline secretion containing prostaglandins, proteins and fructose. h fructose content provides energy to the spermatozoa. These secretions from 60-70% of found in the semen.

Semen = Sperms + Seminal plasma

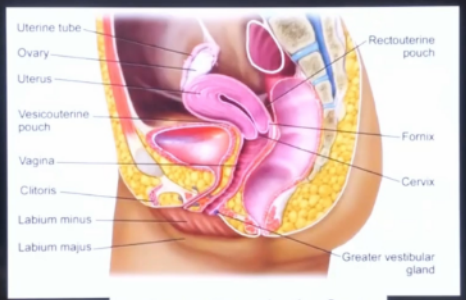
S.V + Prostate + Cowpers

60-70% | 25-30% | 5-10%

Question

Draw a labelled diagram of reproductive system in a human female

Sol.



The Female Reproductive System

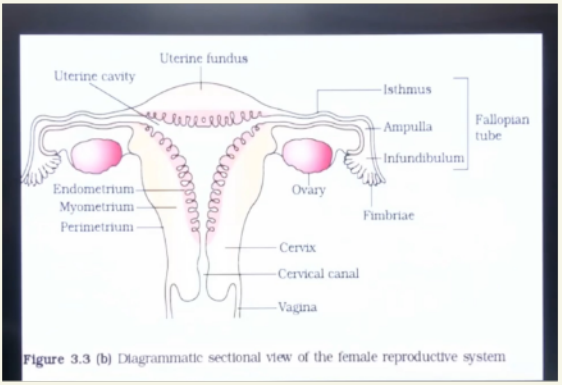


Figure 3.3 (b) Diagrammatic sectional view of the female reproductive system

Question

Mention the differences between spermiogenesis and spermiation (2019, 2 marks)

Sol.

ifference between spermiogenesis and spermiation is

rmiogenesis : It is the process of transforming spermatids into mature spermatozoa or sperms ugh differentiation

miation : Sperm heads are embedded in sertoli cells to obtain nourishement and finally ased from seminiferous tubules by the process called spermiation

Question

Where is acrosome present in humans? Write its function.

Sol.

In humans, the acrosome is present in the anterior portion of the head of human sp

Function Hydrolytic enzymes or sperm lysins present in acrosome help in the penet into egg, during fertilization.

Female reproductive system

Primary sex organ

Ovary

- produce female gamete
- Produce female sex hormones (progesterone and estrogen)

Secondary sex organ

Accessory duct

- *oviduct
- * uterus
- * cervix
- *vagina

External genitalia

- *mons Pubis
- *clitoris
- *labia

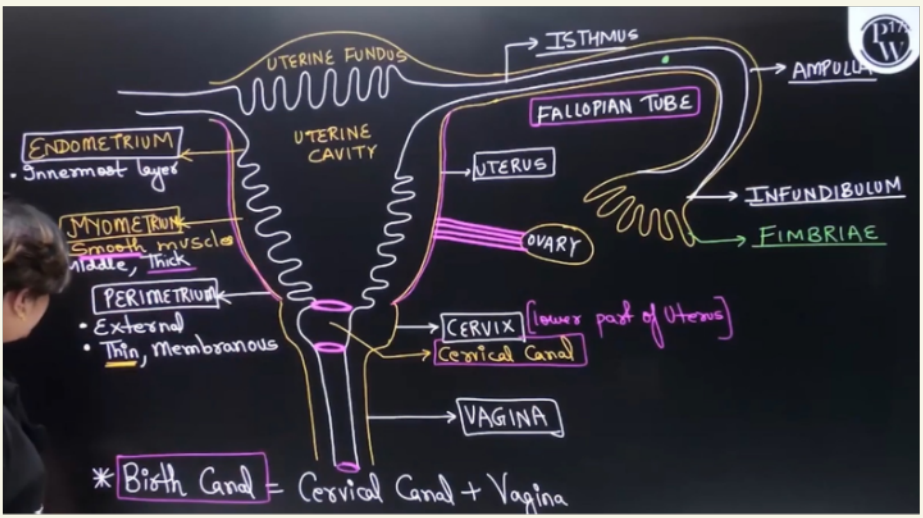
Accessory glands

mammary gland

Majora

* labia minora
* hymen

All of these integrate structurally and functionally to support ovulation, fertilisation, pregnancy, delivery, and child care



Ovary

- one pair
- One on each side
- 2 - 4 cm
- connect to uterus and pelvic wall by ligaments
- Primary sex organ

Fallopian tube

- 10-12 cm
- Three parts

1. Infundibulum- close to the ovary.
   - Finger like projections: fimbriae
2. Ampulla- widest part.
3 isthmus- narrow, last part
   - opens into uterus

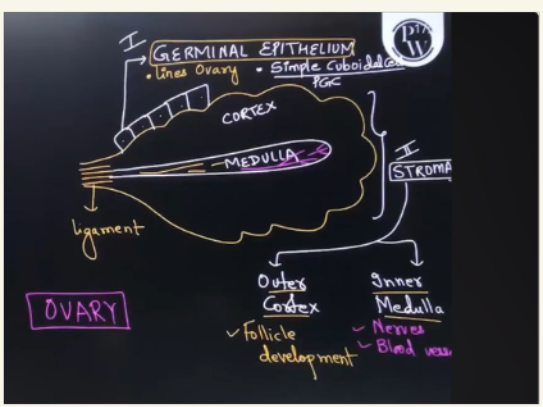
Site of fertilisation: Ampullary Region

Uterus (womb)

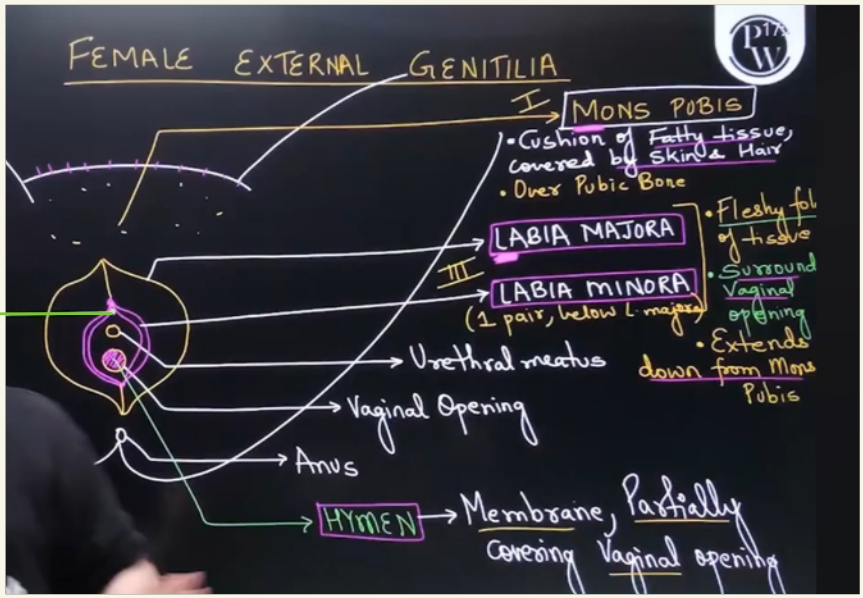
- Inverted Pear shaped
- Single
- Opens into vagina through cervix
- connected to pelvic wall by ligament
- Uterine wall has three Layers

1. Perimetrium
2. Myometrium- strong contraction during childbirth.
3. Endometrium- implantation of blastocyst
   - cyclic changes during menstrual cycle
   - glandular
   - Lines uterine cavity

Structure of ovary



Female external genitalia



Clitoris

Tiny ,finger like

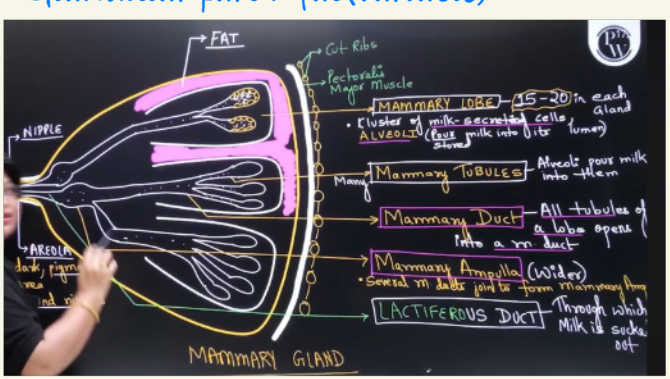
At upper junction of labia minora

Erectile

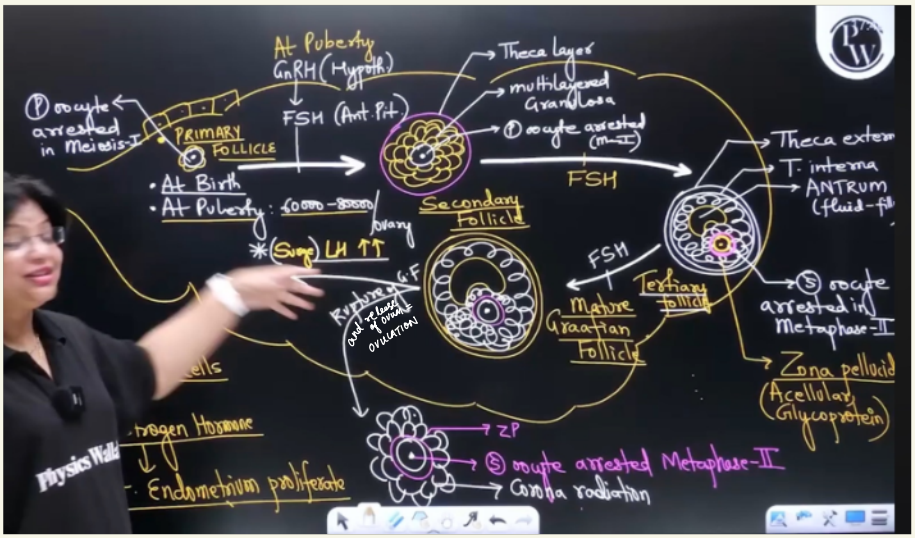
Above urethral opening

Mammary gland

- Character of female mammals
- One pair
- To nourish newborn
- Glandular part+ fat(variable)



Follicular maturation and oogenesis



Corpus luteum- release progesterone
- some amount of estrogen

Oogenesis

Before birth

- multiplication: Oogonia (2n)
- Growth phase: Oogonia (2n) → Primary oocyte (2n)

Since puberty

Primary oocyte (2n) → Secondary oocyte (n) + 1 polar body

Secondary oocyte (n): Arrested in metaphase II

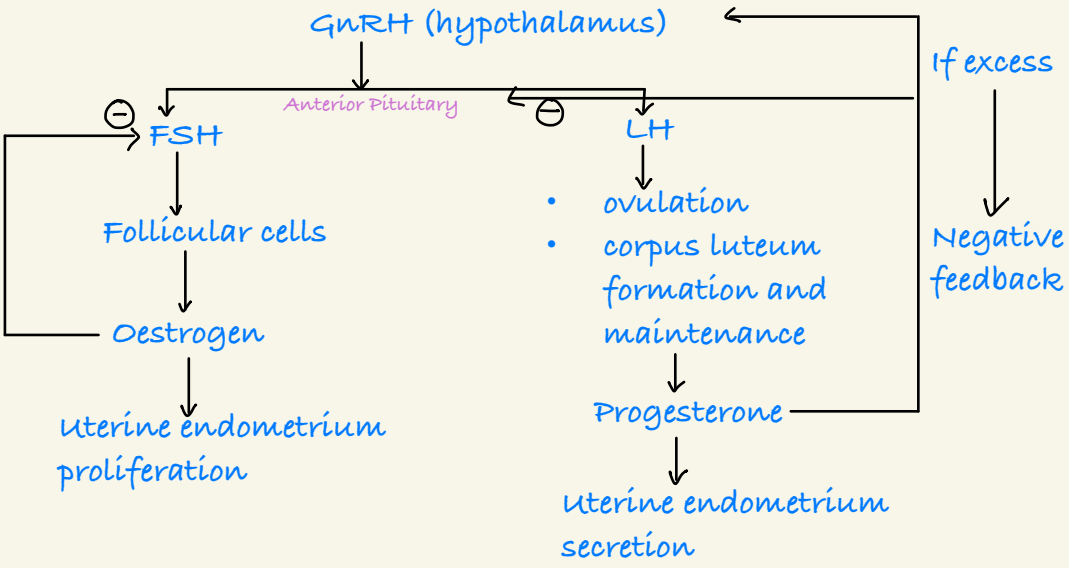
No sperm entry

- Meiosis II not complete
- Meiosis II complete in fallopian tube

Ovum 2nd polar body (Regresses)

Note: Meiosis II completed before fertilisation, but after sperm entry (PYQ)

Hormone regulation in females



Note: HCG acts like LH and maintains pregnancy

Spermatogenesis	OoGenesis
Spermatozoa. Motile Very little cytoplasm	Ovum. Nonmotile Abundance cytoplasm

Question

Draw a sectional view of human ovary showing the different stages of developing follicles, corpus luteum and ovulation

Or

(i) Draw a sectional view of human ovary and label the following parts
- (a) Primary follicle,
- (b) Secondary oocyte
- (c) Graafian follicle
- (d) Corpus luteum

(ii) Name the hormones influencing follicular development of corpus luteum

Sol.
The hormones influencing follicular development of corpus luteum are LH , FSH, estrogen and progesterone



Question

Explain the role of pituitary and sex hormones in the process of spermatogenesis

(2015, 2/5 marks)

OR

Spermatogenesis in human males is a hormone regulated process. Justify

OR

Explain the hormonal regulation of the process of spermatogenesis in hu

Sol.
I. Hormonal control of spermatogenesis in human males is as follows:
II. Gonadotropin Releasing Hormone (GnRH) is released significantly from th during puberty.
III. GnRH stimulates anterior pituitary to secrete gonadotropins, i.e FSH and stimulating Hormone (ICSH)
IV. Luteinising Hormone (LH) acts on leydig cells to stimulate synthe androgens which then stimulate the process of spermatogenesis.
V. Follicle Stimulating Hormone (FSH) acts on sertoli cells and stimulate which help in the process of spermiogenesis.

Question

What happens to corpus luteum in human female if the ovum is (i) fertilized, (ii) not fertilized?

Or

Mention the fate of corpus luteum and its effect on the uterus in the absence of ation of the ovum in human female.

ase of fertilization, the corpus luteum continues secreting progesterone which is required for maintenance of endometrium during pregnancy
the absence of fertilization , the corpus luteum degenerates and gets converted into corpus albicans. Deficiency of progesterone causes disintegration of the endometrium leading to menstruation and thus, a new cycle starts.

Question

Draw a diagram of a mature human sperm. Lab any three parts and write their functions.

Or

Drawn a labelled diagram of a human sperm.

Or

Draw a diagram of human sperm. Label only those parts along w nctions that assist the sperm to reach and gain entry into the female gamete.

Sol.
For structure of a human sperm
The parts that assist sperm to reach and gain entry into female gamete are
Tail : Its wiggling movement helps the sperm to swim in fema reproductive tract
Middle piece : It contains mitochondria that provide energy for sper movement
Head : It contains acrosome loaded with sperm lysins. The help dissolve the layers of female gamete or egg

Question

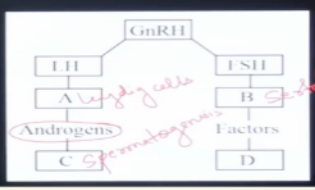
Write the effect of high concentrations of LH on a mature Graafian follicle

Sol.
The high concentration of LH (also known as LH surge) induces the rupture of Graafian follicle, which results in the release of secondary oocyte hence, causing ovulation in females.

Question

Identify A, B, C and D with reference to gametogenesis in humans, in the flow chart given below

Sol.
A – Leyding's cell
B – Sertoli cell
C – Spermatogenesis
D – Spermiogenesis



Question

Differentiate between menarche and menopause are
Ans.

Sol.

Menarche	Menopause
It refers to the beginning of menstrual cycle in human female at puberty	It refers to the stoppage of menstrual cycle at the age of about 45–50 in humans female.
It indicates the start of reproductive phase	It indicates the end of reproductive phase

Question

Describe the roles of pituitary and ovarian hormones during the menstrual cycle in a human female **(2014/2015 , 5 marks)**

Or

Explain the ovarian and uterine events that occur during a menstrual cycle in a human female under the influence of pituitary and ovarian hormones, respectively

OR

the different phases of menstrual cycle and correlate the phase with the different of phases with the different levels of pituitary hormones in a human female.

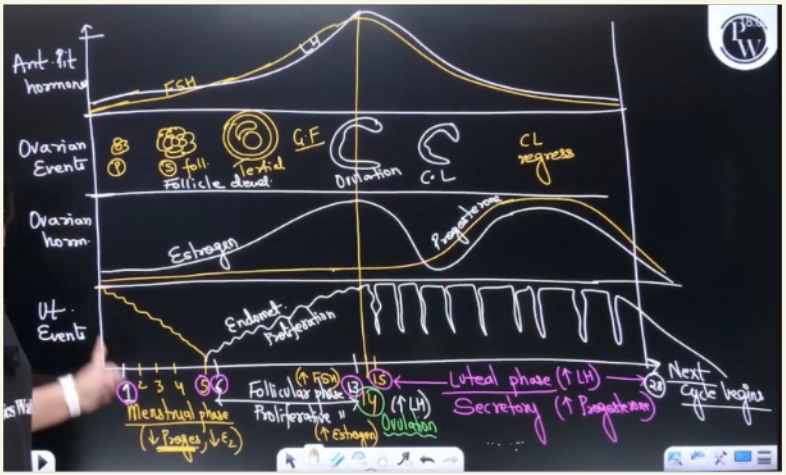
Or

the help of graphical illustration only, show the changes in the levels of the pituitary ormone during menstrual cycle in humans

e of events starting form one menstruation till next in female primates is called ation till next in female primates is called menstrual cycle. It comprises of four phases e regulated by both pituitary (LH and FSH) and ovarian (oestrogen and progesterone) s that affect ovaries and uterus, respectively. The events occurring in a menstrual cycle

Menstrual cycle

- seen in primate mammals (human, apes)
- First menstruation: menarche
- pituitary hormones (FSH, LH) ⟶ Ovarian events.
  Ovarian hormones ⟶ Uterine events



FSH. ⟶ 1 peak. ⟶ Mid cycle.
LH ⟶ 1 peak. ⟶ Midcycle.
Progesterone ⟶ 1 peak. ⟶ Luteal phase.
Oestrogen ⟶ 1 peak. ⟶ Midcycle.

Phases of Menstrual cycle

Menstrual phase - initiated by reduced secretion of LH, progesterone, and Oestrogen. The endometrium breaks down and blood along with unfertilised ovum constitutes normal menstrual flow

Follicular phase/proliferative- (from 6th- 13th day in 28 day cycle) The FSH secreted by anterior pituitary stimulates ovarian follicle to secretes oestrogen. These oestrogen stimulate proliferation of uterine walls as a result of which endometrium gets thickened.

Ovulatory phase- (14th day in a 28 day cycle) Pituitary hormones LH and FSH reach the highest level in the middle of the cycle. Aid secretion of LH causes ovulation, thus including the rupture of graafian follicle to release secondary oocyte.

Luteal phase/secretory- (Day- 15 to 28)

The pituitary hormone LH stimulates the remaining cells of ovarian follicle to develop into corpus luteum. this corpus luteum secrets, large amount of progesterone, and maintains thick endometrium for the implantation of the fertilised ovum during pregnancy. In the absence of fertilisation, the hormone levels are reduced and endometrium disintegrates leading to onset of another menstrual cycle

Fertilisation

Insemination - release of semen by penis into vagina

↓

Motile sperms swim rapidly, enter uterus, and reach ampullary region

↓

Ovum released by ovary, also reaches the ampulla

↓

Sperm comes in contact with Zona pellucida and induces changes in the membrane that block entry of additional sperms

↓

Reaction of acrosome helps sperm enter into cytoplasm of ovum

↓

Induce completion of meiosis II and secondary oocyte

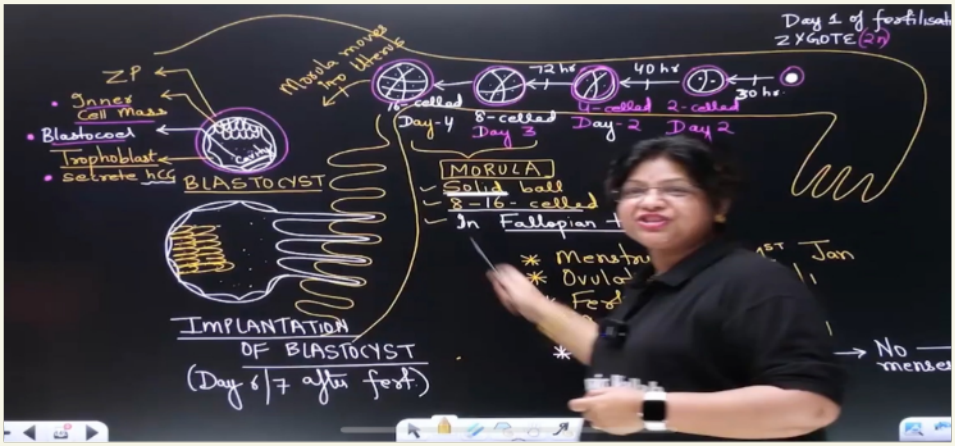
↓

Haploid nucleus of sperm fuses with ovum

↓

Zygote formation

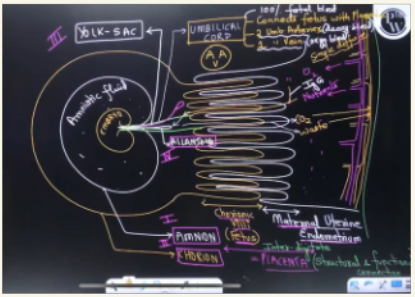
Embryogenesis



Ovulation- day 14 of cycle
Fertilisation - day 14 of cycle
Implantation - day 21 of cycle

Placenta

- structural and functional unit between mother and fetus
- Function: supply oxygen and nutrients to baby
  Removal of carbon dioxide and waste
  Diffusion of IgG antibody into fetus
  Endocrine gland that produces hormone



Placental hormones

- estrogen
- Progesterone
- Relaxin
- hPL - milk secretion

Umbilical cord

Connect fetus to placenta

Has 2 umbilical arteries and 1 umbilical vein

Embryonic development

By the end of 1 month - heart formed

2 month- limbs, digits

3 month- external genitalia, major organ system

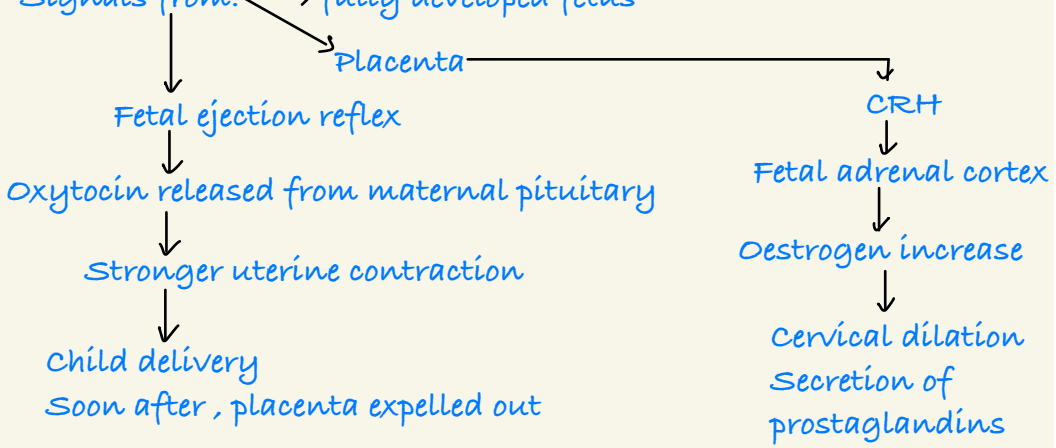
5th month - 1st fetal movement, hair on head

6 month - body hair, eyelashes, eyelids separate

9 month- fetus ready for delivery

Parturition

- delivery of baby/expulsion out of fetus
- Complex neuro endocrine mechanism
- Signals from → fully developed fetus



gestation

Average duration of human pregnancy = 9 months

Lactation

- Milk production starts towards the end of pregnancy to feed the newborn
- Colostrum- initial few days milk, has several antibodies IgA
- Milk secretion - due to prolactin and hPL
- Milk ejection- due to oxytocin